



DAVA RIZKY PRATAMA

0895422957680 | davarizkypratama@students.undip.ac.id | <http://www.linkedin.com/in/davarizkypratama>
Banyumas, Central Java, Indonesia

A sixth-semester undergraduate student in Automation Engineering at Diponegoro University with a strong interest in industrial automation. Combines knowledge from campus studies with real hands-on experience from internships. Former intern in the Maintenance Execution Division at PT. Kilang Pertamina Internasional and the Electrical Department at PT. Sinar Tambang Arthalestari (Semen Bima). Skilled in PLC programming, SCADA, Internet of Things (IoT) integration and instrumentation maintenance, with a commitment to delivering efficient and connected automation solutions.

EDUCATION

Diponegoro University - Semarang, Indonesia August 2023 – Present
Bachelor of Applied Engineering (D4) in Automation Engineering, 3.89/4.00

WORK EXPERIENCES

PT. Kilang Pertamina Internasional Refinery Unit VI - Indramayu, Indonesia January 2026 – February 2026
Internship – Maintenance Execution (Instrument Maintenance Area 1)

- Assisted in the installation of Orifice and Flow Transmitters for Slop Oil Mass Balance at the Treating Unit, executing the full engineering workflow from MOC and P&ID analysis, Orifice Sizing, and Differential Pressure calculations to developing Project Specifications, Hook-up Drawing, Loop Drawing to define electrical signal configuration to the DCS.
- Troubleshoot and resolved discrepancies between field instruments and DCS (Distributed Control System) readings by calculating and applying gain and bias adjustments to ensure data accuracy.
- Assisted technicians in performing inspections, calibrations, and maintenance on various field instruments, including sensors, transmitters, transducers, valves, to ensure plant reliability and safety.

PT. Sinar Tambang Arthalestari (Semen Bima) - Banyumas, Indonesia July 2025 – August 2025
Internship – Electrical Department, Instrumentation Division

- Assisted engineers in conducting inspections and implementing the PDCA (Plan-Do-Check-Action) cycle prior to maintenance activities.
- Performed instrumentation maintenance on various components and industrial machinery.
- Troubleshoot PLC systems to operate and control industrial automation process.

Diponegoro University - Semarang, Indonesia February 2026 – Present
Industrial Instrumentation Laboratory Assistant

- Assisted students in designing and implementing data acquisition systems by integrating LabVIEW with Arduino microcontrollers.
- Facilitated practical experiments demonstrating how real-time acquired data to monitor and control physical processes and automation systems.

Diponegoro University - Semarang, Indonesia February 2026 – Present
Industrial Process Control Laboratory Assistant

- Assisted students using the Festo MPS (Modular Production System) Separating Station to simulate automated industrial sorting processes.
- Instructed students in industrial control programming and logic design using Festo PLCs and CODESYS software.

ORGANIZATIONAL EXPERIENCES

HIMATRO (Automation Engineering Student Association) – Semarang, Indonesia April 2025 – Present
Head of Sports Division

- Led a team to organize the "Automation Futsal Cup," managing 24 participating schools and overseeing event execution.
- Supervised 8 staff members in executing departmental work programs to facilitate student talents.

IWAKMAS UNDIP (Banyumas Student Regional Organization) – Semarang, Indonesia February 2025 – Present
Head of External Diplomacy Division

- Managed external relations and branding activities, building partnerships with other student organizations.
- Led the "Mangrove Planting" project involving 100+ volunteers to support coastal restoration in Semarang.

PROJECT EXPERIENCES

- (2026): IoT Based Temperature and Humidity Monitoring System** – Developed an IoT monitoring system utilizing an ESP microcontroller and a DHT11 sensor to transmit realtime environmental data via the MQTT protocol. Integrated Node-RED as a middleware pipeline to process MQTT topics and execute HTTP requests for secure Firebase database storage, and designed an interactive web-based dashboard using HTML, CSS, and JavaScript for realtime data visualization.

- **(2026): CMMS Development for Voca Water Undip** – Developed a Computerized Maintenance Management System (CMMS) using Google Sheets to digitalize asset management for a university water facility. Designed structured databases to track maintenance schedules, manage work orders, and record equipment history. Streamlined the maintenance workflow by transitioning from manual records to a centralized and easily accessible digital platform.
- **(2026): Automated IoT Hydroponic System** – Developed a realtime control system using an ESP32 microcontroller programmed in C/C++ to regulate water level, temperature, and pH. Implemented a Finite State Machine and PID control algorithms for automated pH dosing using PWM peristaltic pumps. Integrated a Firebase database with a custom HTML, CSS, and JavaScript web dashboard for live data visualization and remote manual control, while utilizing the Telegram API specifically for critical system notifications.
- **(2026): Computer Vision Based PLC Control via Modbus TCP** – Developed a gesture controlled system using MediaPipe computer vision to detect hand movements from a webcam. Used the Modbus TCP protocol to send write coil commands based on finger positions, controlling a Schneider PLC over a local network. Built a custom Python GUI to display live video feeds, system logs, and manual controls, ensuring smooth communication between the software and hardware.
- **(2026): Orifice Flowmeter Design for Light Slop Oil Line** – Designed an orifice flowmeter system for Unit 23/24 at PT Kilang Pertamina Internasional RU VI, including P&ID analysis, flowmeter sizing calculations in accordance with ISO 5167 and API MPMS, piping and material selection based on ASTM, ASME, and ANSI standards, and development of hook-up and loop drawings for mechanical installation and DCS signal integration.
- **(2025): Automatic Water Level Control System** – Designed a dual tank automatic control system using Arduino, ultrasonic sensors, solenoid valves, and LabVIEW based HMI for real-time monitoring.
- **(2025): Automatic Heater** – Developed a temperature based automatic heating control system utilizing an Arduino, a thermocouple sensor, and a Solid State Relay (SSR). Integrated a LabVIEW-based HMI for real-time temperature data acquisition, system monitoring, and bidirectional setpoint control, allowing seamless temperature regulation via both LabVIEW and the Arduino IDE.
- **(2025): Servo Motor Position Control System** – Developed a servo position controller using Arduino and Delphi interface for precise and interactive angle control. Servo angle control with Arduino and Delphi interface for interactive motion control.
- **(2024): DC Motor Speed Control System** – Implemented PID and PWM based DC motor speed control using MATLAB Simulink integrated with Arduino.

SKILLS

- **Soft Skills:** Analytical thinking, problem-solving, and decision-making skills, with the ability to work independently and as part of a team. Possesses tenacity, organizational skills, and a strong sense of initiative.
- **Hard Skills:** C/C++, Python, Microsoft Office (Word, Excel, PowerPoint), JavaScript, HTML, CSS, ESP32, Arduino, STM32, MQTT, HTTP, Modbus, Node-RED, Visual Studio Code, Firebase, MatLab, CX-Programmer, Fluidsim, Proteus, EasyEDA, PSIM, Arduino IDE, LABVIEW, Delphi 7, STM32CubeIDE, STM32CubeMX, EcoStruxure Machine Expert, CtrlX, Solidworks.
- **Languages:** Indonesian, English.